LESSON 2 and LESSON 3

BOOK CHAPTER 22

ELECTRIC FIELDS

Problem 9 (Book chapter 22)

In the adjacent figure shows two charged particles on an *x* axis: $-q = -3.20 \times 10^{-19} C$ at *x* = - 3.00 m and $+q = +3.2 \times 10^{-19} C$ at *x* = + 3.00 m. What are the (a) magnitude and (b) direction (relative to the positive direction of the *x* axis) of the net electric field produced at point P at y = 4 m?

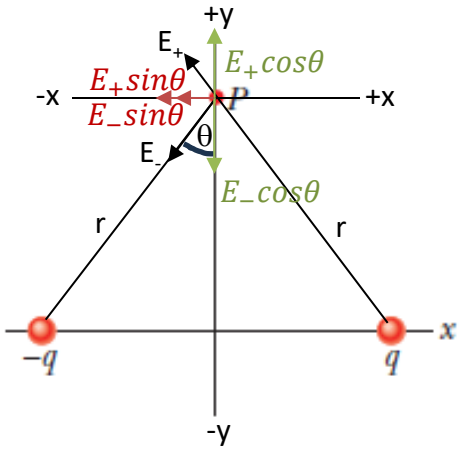
Answer:

$$E_P = E_+ \sin\theta + E_- \sin\theta$$

$$= 2E_+ \sin\theta$$

$$= 2 \times 1.15 \times 10^{-10} \times 0.6$$

$$= 1.38 \times 10^{-10} N/C$$



$$r = \sqrt{3^2 + 4^2} = 5m$$

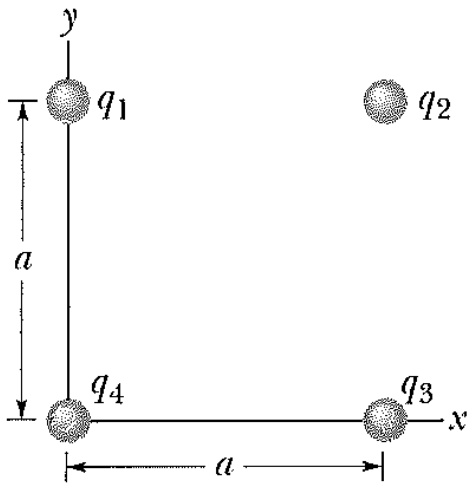
$$\sin\theta = 3/5$$

$$E_+ = E_- = \frac{1}{4\pi\epsilon_o}\frac{q}{r^2} = 9 \times 10^9 \frac{3.2 \times 10^{-19}}{5^2}$$

$$= 1.15 \times 10^{-10} N/C$$

Problem 7 (Book chapter 22)

In the adjacent figure, the four particles form a square of edge length *a* = 5.00 cm and have charges $q_1 = +10\,nC$, $q_2 = -20\,nC$, $q_3 = +20\,nC$, and $q_4 = -10\,nC$. In unit-vector notation, what net electric field do the particles produce at the square's center?



Answer:

The net electric field at the center of the square along x-axis is

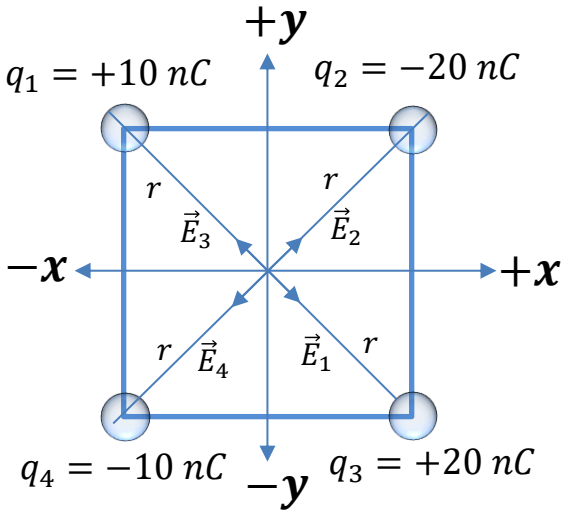
$$E_x = E_2 cos45^0 + E_1 cos45^0 - (E_3 cos45^0 + E_4 cos45^0)$$

Here,

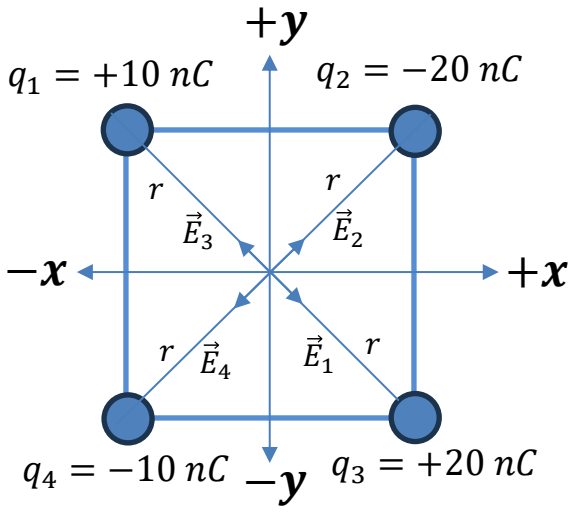
$$E_1 = E_4 = \frac{1}{4\pi\varepsilon_0}\frac{10\times10^{-9}}{r^2} \text{ And } \quad E_3 = E_2 = \frac{1}{4\pi\varepsilon_0}\frac{20\times10^{-9}}{r^2}$$

$$E_x = E_2 cos45^0 + E_1 cos45^0 - E_2 cos45^0 - E_1 cos45^0$$

$$E_x = 0 \text{ N/C}$$



The net electric field at the center of the square along y-axis is



$$E_y = E_2 cos45^0 + E_3 cos45^0 - (E_1 cos45^0 + E_4 cos45^0)$$

$$E_y = E_2 cos45^0 + E_2 cos45^0 - (E_1 cos45^0 + E_1 cos45^0)$$

$$E_y = 2E_2 cos45^0 - 2E_1 cos45^0$$

$$E_y = \frac{2 \times 9 \times 10^9 \times 20 \times 10^{-9} \times 0.707}{r^2} - \frac{2 \times 9 \times 10^9 \times 10 \times 10^{-9} \times 0.707}{r^2}$$

$$r = \sqrt{\frac{a^2}{4} + \frac{a^2}{4}} = \sqrt{\frac{2a^2}{4}}$$

$$r = \frac{a}{\sqrt{2}} = \frac{0.05}{1.414} = 0.0354\ m$$

$$E_y = \frac{127.26}{r^2} = \frac{127.26}{(0.0354)^2} = 101.55 \times 10^3\ N/C$$

$$\vec{E} = E_x \hat{\imath} + E_y \hat{\jmath} = 0 + 101.55 \times 10^3\ \hat{\jmath}$$

$$= 101.55 \times 10^3\ \hat{\jmath}\ \frac{N}{C}$$

Linear charge density:

When charge is distributed along a line (such as a long, thin, charged plastic rod), we use (the Greek letter lambda λ) to represent the charge per unit length known as **linear charge density.**

That is

$$\lambda = \frac{Amount\ of charge\ distributed\ on\ the\ rod}{Length\ of\ the\ rod}$$

[For uniform linear charge density]

The SI unit of λ is Coulomb/meter; simply, we use C/m.

Note:

Some Measures of Electric Charge

Name	Symbol	SI Unit
Charge	q	C
Linear charge density	λ	C/m
Surface charge density	σ	C/m^2
Volume charge density	ρ	C/m^3

Analytical problem:

For charge that is distributed uniformly over a ring, determine the net electric field at a given point on the axis of the ring (at a distance z from the center of the ring).

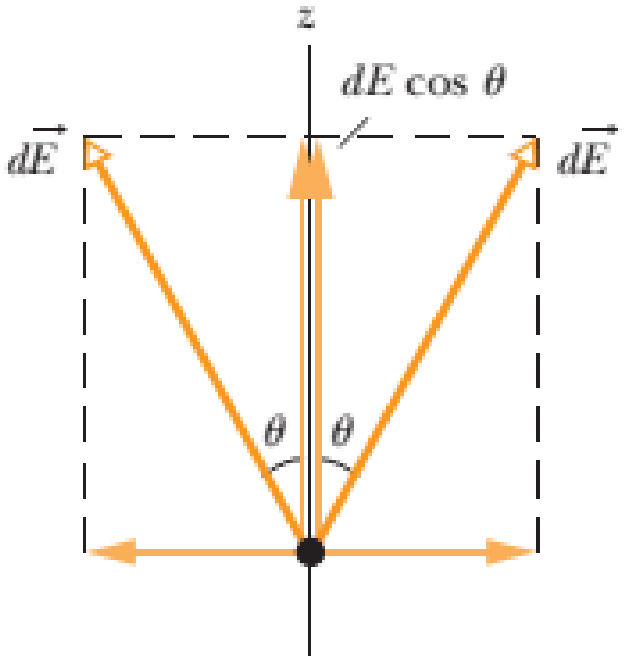
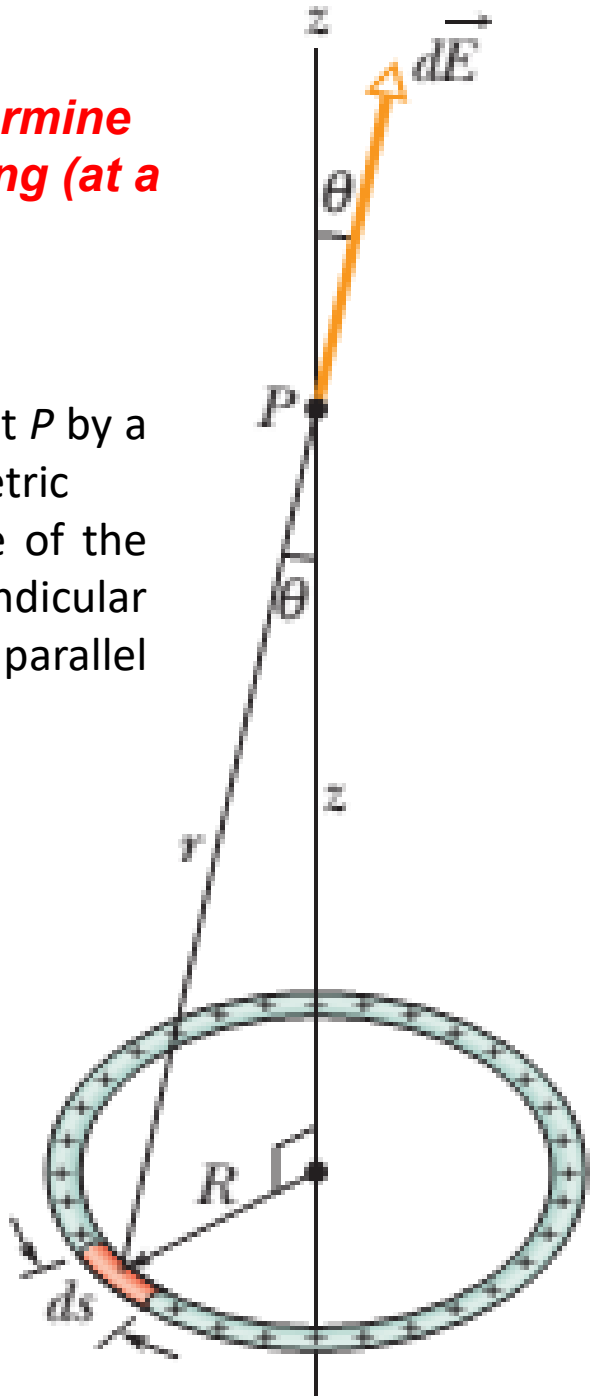


Fig. The electric fields set up at *P* by a charge element and its symmetric partner (on the opposite side of the ring). The components perpendicular to the *z* axis cancel; the parallel components add.



Let *ds* be the (arc) length of any differential element of the ring. Since λ is the charge per unit (arc) length, the element has a charge of magnitude

$$dq = \lambda ds$$

This differential charge (dq) sets up a differential electric field $\mathrm{d}\vec{E}$ at point *P*, which is a distance *r* from the element. Treating the element as a point charge.

Hence, we can write the magnitude of $\mathrm{d}\vec{E}$ as

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{\left(\sqrt{z^2+R^2}\right)^2}$$

Since the components perpendicular to the *z* axis cancel and the parallel components add, the net electric field along z-axis is

$$E_z = E = \int dE\, cos\theta = \int \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{(z^2+R^2)}\left[\frac{z}{\sqrt{z^2+R^2}}\right]$$

$From\ the\ figure,$ we can write

$$cos\theta = \frac{z}{r} = \frac{z}{\sqrt{z^2+R^2}}$$

$$E = \frac{\lambda z}{4\pi\varepsilon_0(z^2+R^2)^{3/2}}\int_{s=0}^{s=2\pi R} ds = \frac{\lambda z}{4\pi\varepsilon_0(z^2+R^2)^{3/2}}[s]_0^{2\pi R}$$

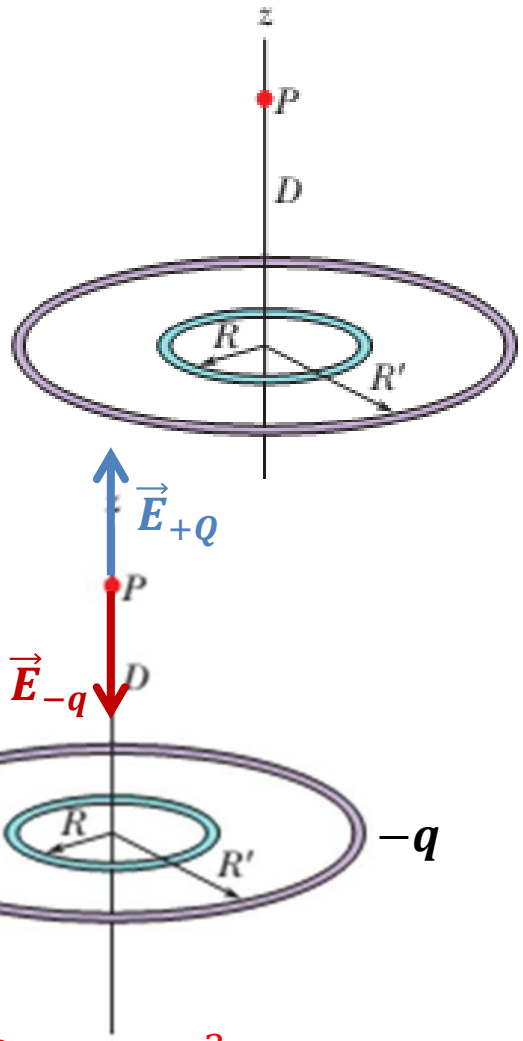
$$E = \frac{\lambda(2\pi R)\, z}{4\pi\varepsilon_0 (z^2 + R^2)^{3/2}} = \frac{qz}{4\pi\varepsilon_0 (z^2 + R^2)^{3/2}}$$

Finally,

$$E = \frac{qz}{4\pi\varepsilon_0 (z^2 + R^2)^{3/2}}$$

Problem 30 (Book chapter 22)

Figure shows two concentric rings, of radii R and $R' = 3R$, that lie on the same plane. Point P lies on the central z axis, at distance $D = 2R$ from the center of the rings. The smaller ring has uniformly distributed charge $+Q$. In terms of Q, what is the uniformly distributed charge on the larger ring if the net electric field at P is zero?



Answer:

According to the statement of the problem,

$$E_{+Q} - E_{-q} = 0$$

$$E_{+Q} = E_{-q}$$

$$\frac{QD}{4\pi\varepsilon_0(D^2 + R^2)^{3/2}} = \frac{qD}{4\pi\varepsilon_0(D^2 + (3R)^2)^{3/2}}$$

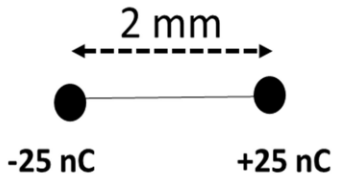
$$\frac{Q}{(4R^2 + R^2)^{3/2}} = \frac{q}{(4R^2 + 9R^2)^{3/2}}$$

$$q = \frac{Q(13R^2)^{\frac{3}{2}}}{(5R^2)^{\frac{3}{2}}} = \left(\frac{13}{5}\right)^{\frac{3}{2}} Q = 4.19Q$$

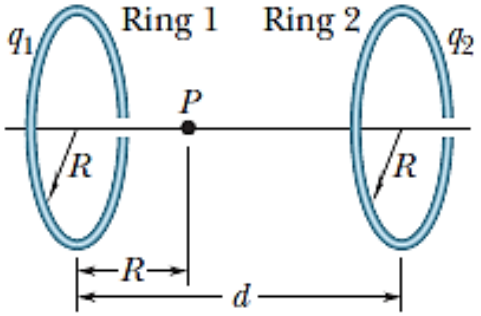
That is $\boxed{q = -4.19Q}$

Let's try !!!

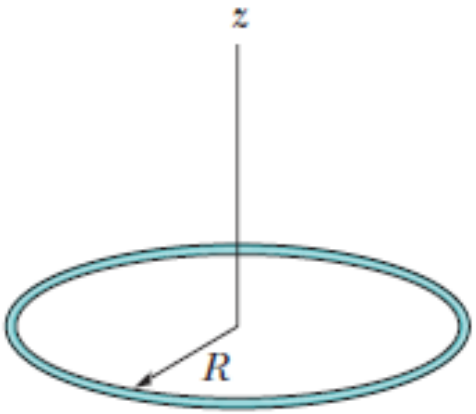
1. Calculate electric dipole moment for the adjacent figure.



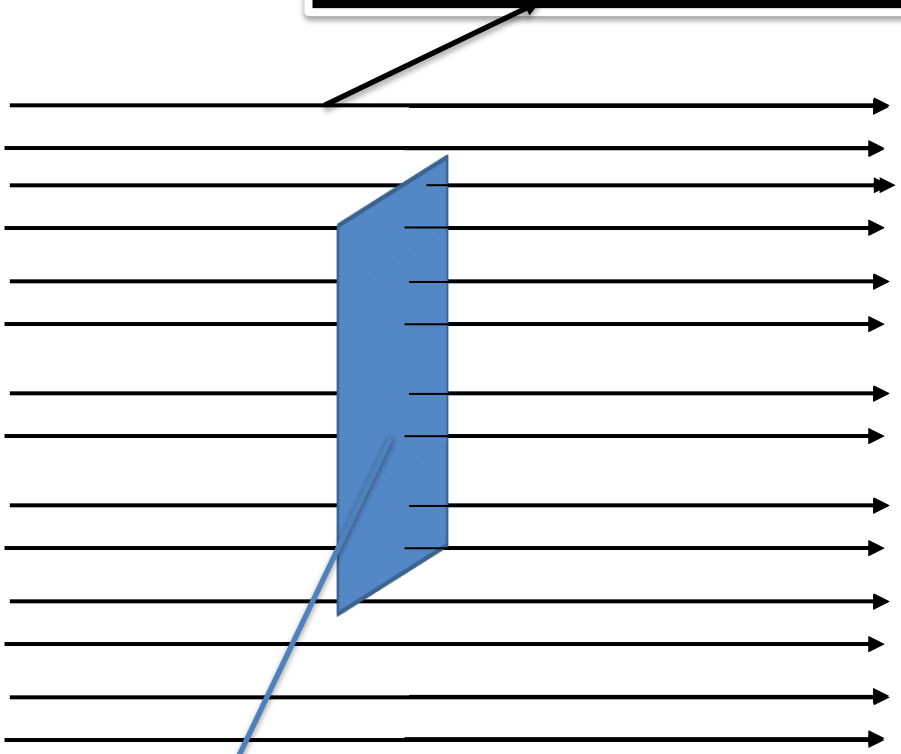
2. [Problem 23]: The adjacent Figure shows two parallel nonconducting rings with their central axes along a common line. Ring 1 has uniform charge q_1 and radius R; ring 2 has uniform charge q_2 and the same radius R. The rings are separated by distance d = 3.00R. The net electric field at point P on the common line, at distance R from ring 1, is zero. What is the ratio q_1/q_2?



3. [Problem 24]: A thin nonconducting rod with a uniform distribution of positive charge *Q* is bent into a complete circle of radius *R.* The central perpendicular axis through the ring is a *z* axis, with the origin at the center of the ring. What is the magnitude of the electric field due to the rod at (a) $z = 0$ and (b) $z = \infty$? (c) In terms of *R*, at what positive value of *z* is that magnitude maximum?



Electric flux



$2\vec{E}$

Look at the no. of field lines of field strength $\vec{E}$ penetrating the surface.

What happens if we increase the field strength to $2\vec{E}$?

What if we increase the surface area to 2A?

Electric Flux:

The electric flux Φ through a surface is the amount of electric field that pierces the surface.

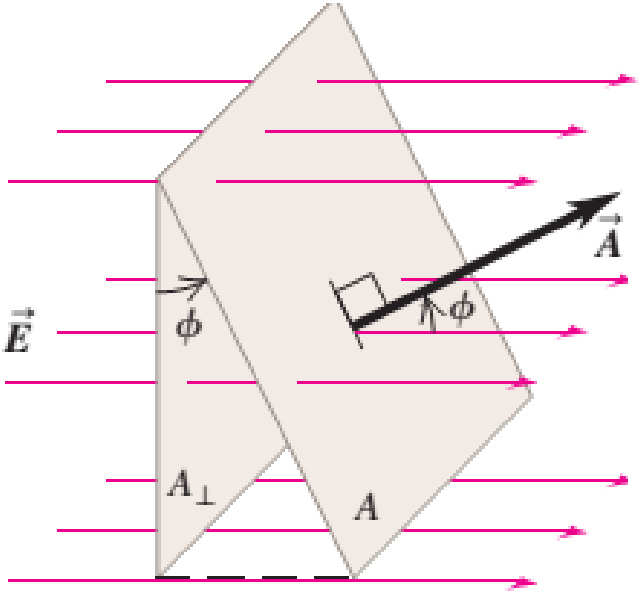
Electric Flux for uniform $\vec{E}$ and through a flat surface (open plane):

The electric flux φ through the flat surface equals the scalar product of the electric field $\vec{E}$ and the area vector $\vec{A}$.

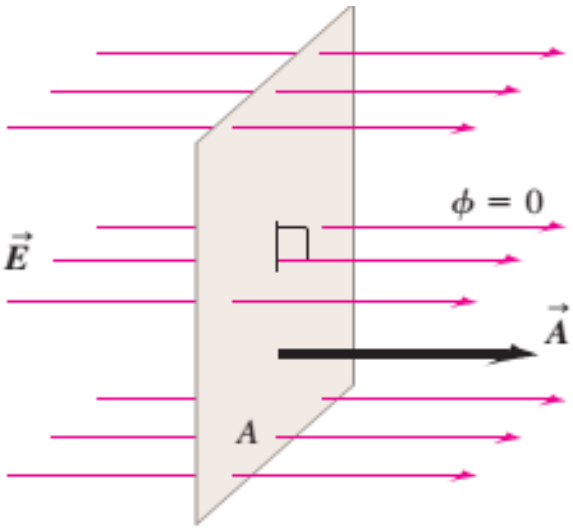
That is, $\Phi = \vec{E}.\vec{A}$

We can represent the direction of a vector area $\vec{A}$ by using a unit vector $\hat{n}$ perpendicular to the area: $\hat{n}$ stands for normal." Then $\vec{A} = A\hat{n}$

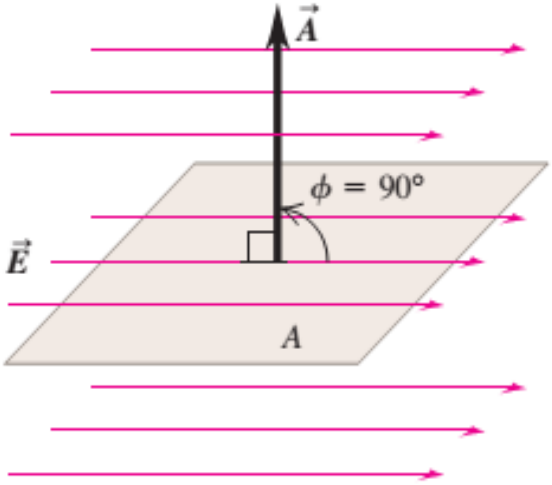
We can write as $\varphi = EAcos\phi$



$\varphi = EAcos0^0 = EA$



$\varphi = EAcos90^0 = 0 \frac{N-m^2}{C}$



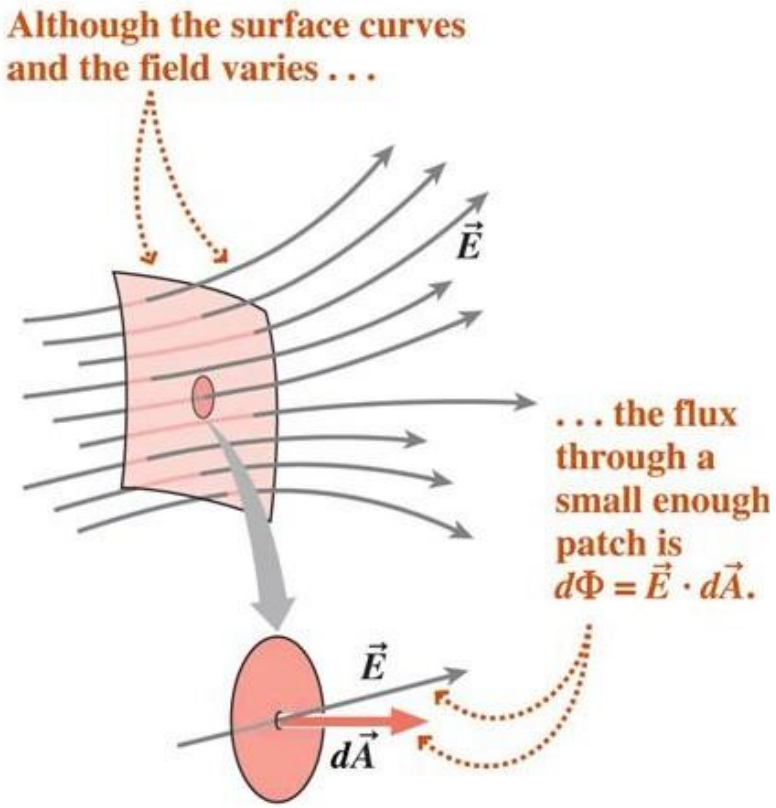
Electric Flux for nonuniform $\vec{E}$ and through a curved surface:

When the surface is curved and field is nonuniform, we calculate the flux by dividing the surface into small (differential) patches (areas) dA, so that each differential area is essentially flat and field is uniform over each patch.

The total flux through the surface is given by

$$\Phi = \int \vec{E}.d\vec{A}$$

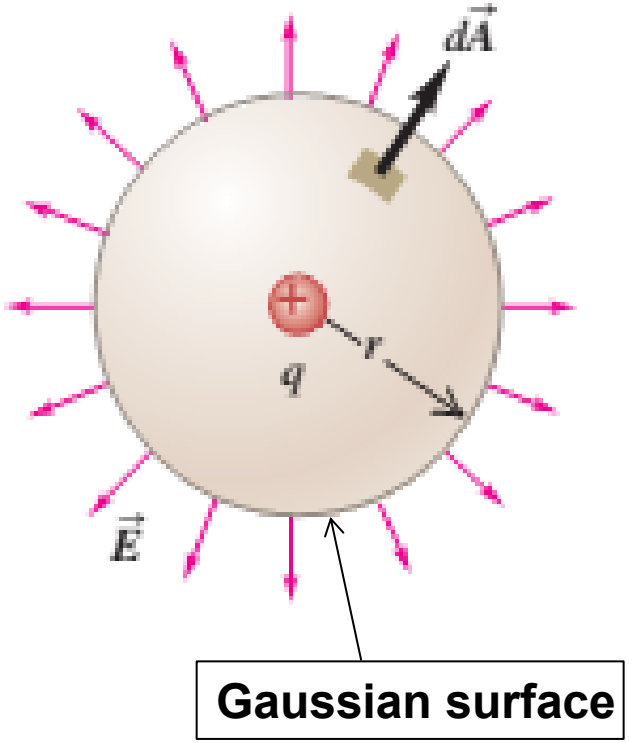
where the integration is carried out over the surface.



Gauss's law:

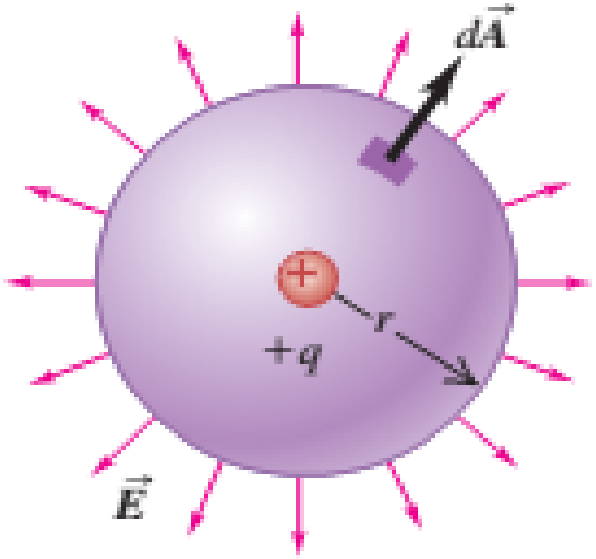
Gauss's law states that the total electric flux(Φ) through any closed surface, known as Gaussian surface, (a surface enclosing a definite volume) is proportional to the total (net) electric charge(q) inside the surface. That is

$$\boldsymbol{\Phi = \frac{q}{\varepsilon_0}}$$

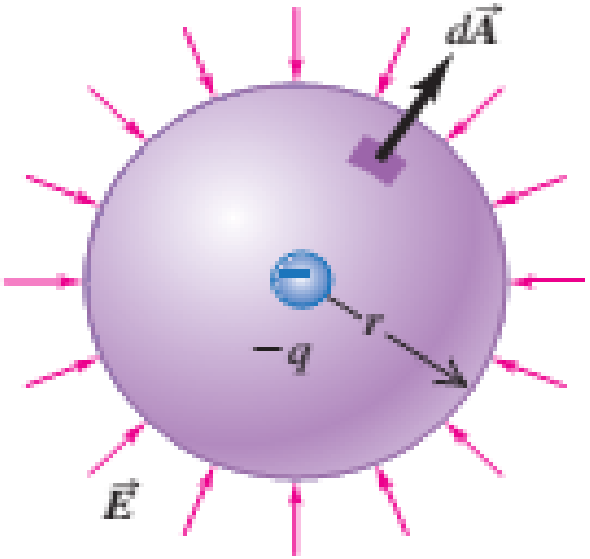


We can write the above form as $\varepsilon_0 \Phi = q$

$$\varepsilon_0 \oint \vec{E}.d\vec{A} = q$$



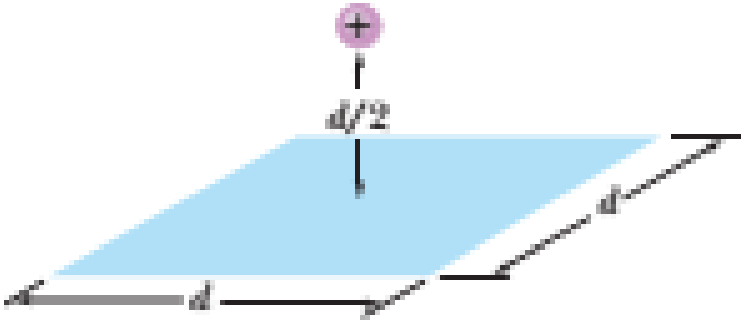
Gaussian surface around positive charge:
positive (outward) flux



Gaussian surface around negative charge:
negative (inward) flux

Problem 5 (book chapter 23)

A proton is at a distance *d/2* directly above the center of a square of side *d*. What is the magnitude of the electric flux through the square? *(Hint:* Think of the square as one face of a cube with edge *d.)*



Answer:

Consider a cube consisting with six square faces; each of edge d. Thus, the proton is enclosed by the cubical Gaussian surface.

Electric flux through the cubical Gaussian surface is

$$\Phi = \frac{q}{\varepsilon_0} = \frac{e^+}{\varepsilon_0} = \frac{1.6 \times 10^{-19}}{8.854 \times 10^{-12}} = 0.18 \times 10^{-7} = 18 \times 10^{-9} \ \frac{N.m^2}{C}$$

Electric flux through a face (a square) of a cube is

$$\Phi_s = \frac{18 \times 10^{-9}}{6} = 3 \times 10^{-9} \ \frac{N.m^2}{C}$$

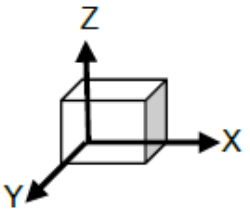
Let's try !!!

A. Calculation of Electric flux:

1. A surface has the area vector $\vec{A} = (2\hat{\imath} + 3\hat{\jmath})\boldsymbol{m^2}$. What is the flux of an electric field through it, if the field is (a) $\vec{E} = 4\hat{\imath}\, N/C$ and (b) $\vec{E} = 5\hat{k}\, N/C$

B. Calculation of Electric flux from Gauss's Law

2. A point charge of 1.8 μC is placed at the center of a cubical Gaussian surface 55 cm on edge. What is the electric flux through the surface?



3. A point charge causes an electric flux of – 750 N m^2/C to pass through a spherical Gaussian surface of 10.0 cm radius centered on the charge. (a) If the radius of the Gaussian surface were doubled, how much flux would pass through the surface. (b) What is the value of the point charge?

4. A point charge q_1 = 5.0 nC is located at the origin, and a second point charge q_2 = - 3.0 nC in on the x axis at x = 1.0 m. What is the total electric flux due to these two point charges through a spherical surface centered at the origin and with radius a) 0.5 m, b) 1.5 m, and c) 2.5 m?

THANK YOU